8 THE VARIATIONAL PRINCIPLE

8.1 THEORY

Suppose you want to calculate the ground state energy, E_{gs}, for a system described by the Hamiltonian H, but you are unable to solve the (time-independent) Schrödinger equation. The **variational principle** will get you an *upper bound* for E_{gs}, which is sometimes all you need, and often, if you're clever about it, very close to the exact value. Here's how it works: Pick *any normalized function* ψ *whatsoever*; I claim that

$$\boxed{E_{\text{gs}} \leq \langle \psi | H | \psi \rangle \equiv \langle H \rangle .} \tag{8.1}$$

That is, the expectation value of H, in the (presumably incorrect) state ψ is certain to *overestimate* the ground state energy. Of course, if ψ just happens to be one of the *excited* states, then *obviously* $\langle H \rangle$ exceeds E_{gs}; the point is that the same holds for any ψ whatsoever.

Proof: Since the (unknown) eigenfunctions of H form a complete set, we can express ψ as a linear combination of them:[1]

$$\psi = \sum_n c_n \psi_n, \quad \text{with } H\psi_n = E_n \psi_n.$$

Since ψ is normalized,

$$1 = \langle \psi | \psi \rangle = \left\langle \sum_m c_m \psi_m \middle| \sum_n c_n \psi_n \right\rangle = \sum_m \sum_n c_m^* c_n \langle \psi_m | \psi_n \rangle = \sum_n |c_n|^2,$$

(assuming the eigenfunctions themselves have been orthonormalized: $\langle \psi_m | \psi_n \rangle = \delta_{mn}$). Meanwhile,

$$\langle H \rangle = \left\langle \sum_m c_m \psi_m \middle| H \sum_n c_n \psi_n \right\rangle = \sum_m \sum_n c_m^* E_n c_n \langle \psi_m | \psi_n \rangle = \sum_n E_n |c_n|^2.$$

But the ground state energy is, by definition, the *smallest* eigenvalue, $E_{\text{gs}} \leq E_n$, and hence

$$\langle H \rangle \geq E_{\text{gs}} \sum_n |c_n|^2 = E_{\text{gs}},$$

which is what we were trying to prove.

[1] If the Hamiltonian admits scattering states, as well as bound states, then we'll need an integral as well as a sum, but the argument is unchanged.

This is hardly surprising. After all, ψ might be the *actual* wave function (at, say, $t = 0$). If you measured the particle's energy you'd be certain to get one of the eigenvalues of H, the smallest of which is E_{gs}, so the *average* of multiple measurements ($\langle H \rangle$) cannot be lower than E_{gs}.

Example 8.1

Suppose we want to find the ground state energy for the one-dimensional harmonic oscillator:

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2.$$

Of course, we already know the *exact* answer in this case (Equation 2.62): $E_{gs} = (1/2)\,\hbar\omega$; but this makes it a good test of the method. We might pick as our "trial" wave function the gaussian,

$$\psi(x) = Ae^{-bx^2}, \tag{8.2}$$

where b is a constant, and A is determined by normalization:

$$1 = |A|^2 \int_{-\infty}^{\infty} e^{-2bx^2}\,dx = |A|^2\sqrt{\frac{\pi}{2b}} \;\Rightarrow\; A = \left(\frac{2b}{\pi}\right)^{1/4}. \tag{8.3}$$

Now

$$\langle H \rangle = \langle T \rangle + \langle V \rangle, \tag{8.4}$$

where, in this case,

$$\langle T \rangle = -\frac{\hbar^2}{2m}|A|^2 \int_{-\infty}^{\infty} e^{-bx^2}\frac{d^2}{dx^2}\left(e^{-bx^2}\right)dx = \frac{\hbar^2 b}{2m}, \tag{8.5}$$

and

$$\langle V \rangle = \frac{1}{2}m\omega^2|A|^2 \int_{-\infty}^{\infty} e^{-2bx^2}x^2\,dx = \frac{m\omega^2}{8b},$$

so

$$\langle H \rangle = \frac{\hbar^2 b}{2m} + \frac{m\omega^2}{8b}. \tag{8.6}$$

According to Equation 8.1, this exceeds E_{gs} *for any* b; to get the *tightest* bound, let's *minimize* $\langle H \rangle$:

$$\frac{d}{db}\langle H \rangle = \frac{\hbar^2}{2m} - \frac{m\omega^2}{8b^2} = 0 \;\Rightarrow\; b = \frac{m\omega}{2\hbar}.$$

Putting this back into $\langle H \rangle$, we find

$$\langle H \rangle_{\min} = \frac{1}{2}\hbar\omega. \tag{8.7}$$

In this case we hit the ground state energy right on the nose—because (obviously) I "just happened" to pick a trial function with precisely the form of the *actual* ground state (Equation 2.60). But the gaussian is very easy to work with, so it's a popular trial function, even when it bears little resemblance to the true ground state.

Example 8.2

Suppose we're looking for the ground state energy of the delta function potential:

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - \alpha\delta(x).$$

Again, we already know the exact answer (Equation 2.132): $E_{gs} = -m\alpha^2/2\hbar^2$. As before, we'll use a gaussian trial function (Equation 8.2). We've already determined the normalization, and calculated $\langle T\rangle$; all we need is

$$\langle V\rangle = -\alpha|A|^2\int_{-\infty}^{\infty} e^{-2bx^2}\delta(x)\,dx = -\alpha\sqrt{\frac{2b}{\pi}}.$$

Evidently

$$\langle H\rangle = \frac{\hbar^2 b}{2m} - \alpha\sqrt{\frac{2b}{\pi}}, \tag{8.8}$$

and we know that this exceeds E_{gs} for all b. Minimizing it,

$$\frac{d}{db}\langle H\rangle = \frac{\hbar^2}{2m} - \frac{\alpha}{\sqrt{2\pi b}} = 0 \;\Rightarrow\; b = \frac{2m^2\alpha^2}{\pi\hbar^4}.$$

So

$$\langle H\rangle_{\min} = -\frac{m\alpha^2}{\pi\hbar^2}, \tag{8.9}$$

which is indeed somewhat higher than E_{gs}, since $\pi > 2$.

I said you can use *any* (normalized) trial function ψ whatsoever, and this is true in a sense. However, for *discontinuous* functions it takes some fancy footwork to assign a sensible meaning to the second derivative (which you need, in order to calculate $\langle T\rangle$). Continuous functions with kinks in them are fair game, however, as long as you are careful; the next example shows how to handle them.[2]

Example 8.3

Find an upper bound on the ground state energy of the one-dimensional infinite square well (Equation 2.22), using the "triangular" trial wave function (Figure 8.1):[3]

$$\psi(x) = \begin{cases} Ax, & 0 \le x \le a/2, \\ A(a-x), & a/2 \le x \le a, \\ 0, & \text{otherwise,} \end{cases} \tag{8.10}$$

where A is determined by normalization:

$$1 = |A|^2\left[\int_0^{a/2} x^2\,dx + \int_{a/2}^{a}(a-x)^2\,dx\right] = |A|^2\frac{a^3}{12} \;\Rightarrow\; A = \frac{2}{a}\sqrt{\frac{3}{a}}. \tag{8.11}$$

[2] For a collection of interesting examples see W. N. Mei, *Int. J. Math. Educ. Sci. Tech.* **30**, 513 (1999).

[3] There is no point in trying a function (such as the gaussian) that extends outside the well, because you'll get $\langle V\rangle = \infty$, and Equation 8.1 tells you nothing.

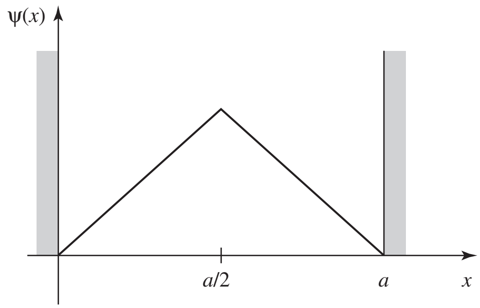

Figure 8.1: Triangular trial wave function for the infinite square well (Equation 8.10).

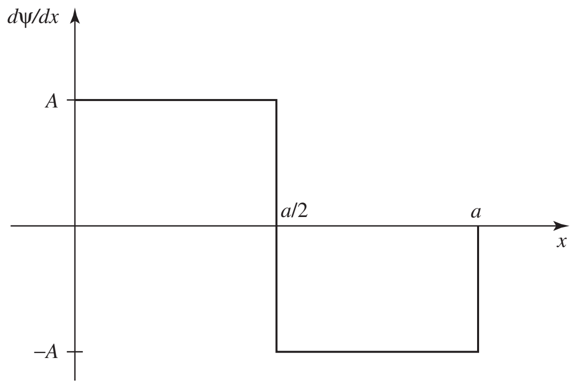

Figure 8.2: Derivative of the wave function in Figure 8.1.

In this case

$$\frac{d\psi}{dx} = \begin{cases} A, & 0 \leq x \leq a/2, \\ -A, & a/2 \leq x \leq a, \\ 0, & \text{otherwise,} \end{cases}$$

as indicated in Figure 8.2. Now, the derivative of a step function is a delta function (see Problem 2.23(b)):

$$\frac{d^2\psi}{dx^2} = A\delta(x) - 2A\delta(x - a/2) + A\delta(x - a), \tag{8.12}$$

and hence

$$\begin{aligned} \langle H \rangle &= -\frac{\hbar^2 A}{2m} \int \left[\delta(x) - 2\delta(x - a/2) + \delta(x - a)\right] \psi(x)\, dx \\ &= -\frac{\hbar^2 A}{2m} \left[\psi(0) - 2\psi(a/2) + \psi(a)\right] = \frac{\hbar^2 A^2 a}{2m} = \frac{12\hbar^2}{2ma^2}. \end{aligned} \tag{8.13}$$

The exact ground state energy is $E_{\text{gs}} = \pi^2\hbar^2/2ma^2$ (Equation 2.30), so the theorem works $\left(12 > \pi^2\right)$.

Alternatively, you can exploit the hermiticity of $\hat{p}$:

$$\langle H\rangle = \frac{1}{2m}\left\langle \hat{p}^2\right\rangle = \frac{1}{2m}\langle \hat{p}\psi \mid \hat{p}\psi\rangle = \frac{1}{2m}\int_0^a \left(-i\hbar\frac{d\psi}{dx}\right)^* \left(-i\hbar\frac{d\psi}{dx}\right) dx$$

$$= \frac{\hbar^2}{2m}\left[\int_0^{a/2} (A)^2\, dx + \int_{a/2}^{a} (-A)^2\, dx\right] = \frac{\hbar^2}{2m}A^2 a = \frac{12\hbar^2}{2ma^2}. \tag{8.14}$$

The variational principle is extraordinarily powerful, and embarrassingly easy to use. What a physical chemist does, to find the ground state energy of some complicated molecule, is write down a trial wave function with a large number of adjustable parameters, calculate $\langle H\rangle$, and tweak the parameters to get the lowest possible value. Even if ψ has little resemblance to the true wave function, you often get miraculously accurate values for E_{gs}. Naturally, if you have some way of guessing a *realistic* ψ, so much the better. The only *trouble* with the method is that you never know for sure how close you are to the target—all you can be *certain* of is that you've got an *upper bound*.[4] Moreover, as it stands the technique applies only to the ground state (see, however, Problem 8.4).[5]

* **Problem 8.1** Use a gaussian trial function (Equation 8.2) to obtain the lowest upper bound you can on the ground state energy of (a) the linear potential: $V(x) = \alpha|x|$; (b) the quartic potential: $V(x) = \alpha x^4$.

** **Problem 8.2** Find the best bound on E_{gs} for the one-dimensional harmonic oscillator using a trial wave function of the form

$$\psi(x) = \frac{A}{x^2 + b^2},$$

where A is determined by normalization and b is an adjustable parameter.

Problem 8.3 Find the best bound on E_{gs} for the delta function potential $V(x) = -\alpha\delta(x)$, using a triangular trial function (Equation 8.10, only centered at the origin). This time a is an adjustable parameter.

Problem 8.4

(a) Prove the following corollary to the variational principle: If $\langle\psi|\psi_{\mathrm{gs}}\rangle = 0$, then $\langle H\rangle \ge E_{\mathrm{fe}}$, where E_{fe} is the energy of the first excited state. *Comment:* If we can find a trial function that is orthogonal to the exact ground state, we can get

[4] In practice this isn't much of a limitation, and there are sometimes ways of estimating the accuracy. The binding energy of helium has been calculated to many significant digits in this way (see for example G. W. Drake *et al.*, *Phys. Rev. A* **65**, 054501 (2002), or Vladimir I. Korobov, *Phys. Rev. A* **66**, 024501 (2002).

[5] For a systematic extension of the variational principle to the calculation of excited state energies see, for example, Linus Pauling and E. Bright Wilson, *Introduction to Quantum Mechanics, With Applications to Chemistry*, McGraw-Hill, New York (1935, paperback edition 1985), Section 26.

an upper bound on the *first excited state*. In general, it's difficult to be sure that ψ is orthogonal to ψ_{gs}, since (presumably) we don't *know* the latter. However, if the potential $V(x)$ is an *even* function of x, then the ground state is likewise even, and hence any *odd* trial function will automatically meet the condition for the corollary.[6]

(b) Find the best bound on the first excited state of the one-dimensional harmonic oscillator using the trial function

$$\psi(x) = Axe^{-bx^2}.$$

Problem 8.5 Using a trial function of your own devising, obtain an upper bound on the ground state energy for the "bouncing ball" potential (Equation 2.185), and compare it with the exact answer (Problem 2.59): $E_{gs} = 2.33811\left(mg^2\hbar^2/2\right)^{1/3}$.

Problem 8.6

(a) Use the variational principle to prove that first-order non-degenerate perturbation theory always *overestimates* (or at any rate never *underestimates*) the ground state energy.

(b) In view of (a), you would expect that the *second*-order correction to the ground state is always negative. Confirm that this is indeed the case, by examining Equation 7.15.

8.2 THE GROUND STATE OF HELIUM

The helium atom (Figure 8.3) consists of two electrons in orbit around a nucleus containing two protons (also some neutrons, which are irrelevant to our purpose). The Hamiltonian for this system (ignoring fine structure and smaller corrections) is:

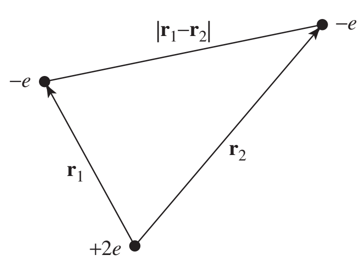

Figure 8.3: The helium atom.

[6] You can extend this trick to other symmetries. Suppose there is a Hermitian operator A such that $[A, H] = 0$. The ground state (assuming it is nondegenerate) must be an eigenstate of A; call the eigenvalue λ: $A\,\psi_{gs} = \lambda\,\psi_{gs}$. If you choose a variational function ψ that is an eigenstate of A with a *different* eigenvalue: $A\,\psi = \nu\,\psi$ with $\lambda \neq \nu$, you can be certain that ψ and ψ_{gs} are orthogonal (see Section 3.3). For an application see Problem 8.20.

$$H = -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) - \frac{e^2}{4\pi\epsilon_0}\left(\frac{2}{r_1} + \frac{2}{r_2} - \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}\right). \tag{8.15}$$

Our problem is to calculate the ground state energy, E_{gs}. Physically, this represents the amount of energy it would take to strip off both electrons. (Given E_{gs} it is easy to figure out the "ionization energy" required to remove a *single* electron—see Problem 8.7.) The ground state energy of helium has been measured to great precision in the laboratory:

$$E_{gs} = -78.975 \text{ eV} \quad \text{(experimental)}. \tag{8.16}$$

This is the number we would like to reproduce theoretically.

It is curious that such a simple and important problem has no known exact solution.[7] The trouble comes from the electron–electron repulsion,

$$V_{ee} = \frac{e^2}{4\pi\epsilon_0}\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}. \tag{8.17}$$

If we ignore this term altogether, H splits into two independent hydrogen Hamiltonians (only with a nuclear charge of $2e$, instead of e); the exact solution is just the product of hydrogenic wave functions:

$$\psi_0(\mathbf{r}_1, \mathbf{r}_2) \equiv \psi_{100}(\mathbf{r}_1)\psi_{100}(\mathbf{r}_2) = \frac{8}{\pi a^3}e^{-2(r_1+r_2)/a}, \tag{8.18}$$

and the energy is $8E_1 = -109$ eV (Equation 5.42).[8] This is a long way from -79 eV, but it's a start.

To get a better approximation for E_{gs} we'll apply the variational principle, using ψ_0 as the trial wave function. This is a particularly convenient choice because it's an eigenfunction of *most* of the Hamiltonian:

$$H\psi_0 = (8E_1 + V_{ee})\,\psi_0. \tag{8.19}$$

Thus

$$\langle H \rangle = 8E_1 + \langle V_{ee} \rangle, \tag{8.20}$$

where[9]

$$\langle V_{ee} \rangle = \left(\frac{e^2}{4\pi\epsilon_0}\right)\left(\frac{8}{\pi a^3}\right)^2 \int \frac{e^{-4(r_1+r_2)/a}}{|\mathbf{r}_1 - \mathbf{r}_2|}\, d^3\mathbf{r}_1\, d^3\mathbf{r}_2. \tag{8.21}$$

I'll do the $\mathbf{r}_2$ integral first; for this purpose $\mathbf{r}_1$ is fixed, and we may as well orient the $\mathbf{r}_2$ coordinate system so that the polar axis lies along $\mathbf{r}_1$ (see Figure 8.4). By the law of cosines,

$$|\mathbf{r}_1 - \mathbf{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2\cos\theta_2}, \tag{8.22}$$

[7] There do exist exactly soluble three-body problems with many of the qualitative features of helium, but using non-Coulombic potentials (see Problem 8.24).

[8] Here a is the ordinary Bohr radius and $E_n = -13.6/n^2$ eV is the nth Bohr energy; recall that for a nucleus with atomic number Z, $E_n \to Z^2 E_n$ and $a \to a/Z$ (Problem 4.19). The spin configuration associated with Equation 8.18 will be antisymmetric (the singlet).

[9] You can, if you like, interpret Equation 8.21 as first-order perturbation theory, with $H' = V_{ee}$ (Problem 7.56(a)). However, I regard this as a misuse of the method, since the perturbation is comparable in size to the unperturbed potential. I prefer, therefore, to think of it as a variational calculation, in which we are looking for a rigorous upper bound on E_{gs}.

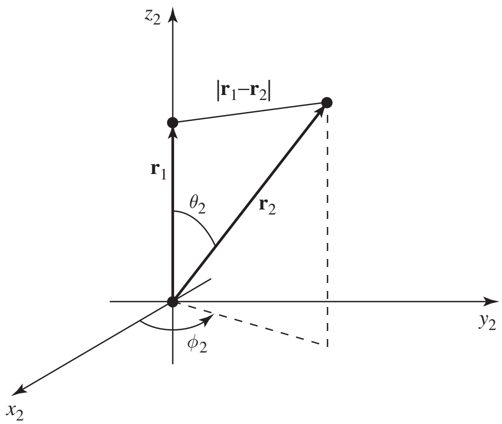

Figure 8.4: Choice of coordinates for the $\mathbf{r}_2$-integral (Equation 8.21).

and hence

$$I_2 \equiv \int \frac{e^{-4r_2/a}}{|\mathbf{r}_1 - \mathbf{r}_2|} d^3\mathbf{r}_2 = \int \frac{e^{-4r_2/a}}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta_2}} r_2^2 \sin\theta_2 \, dr_2 \, d\theta_2 \, d\phi_2. \tag{8.23}$$

The ϕ_2 integral is trivial (2π); the θ_2 integral is

$$\int_0^\pi \frac{\sin\theta_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta_2}} d\theta_2 = \left. \frac{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta_2}}{r_1 r_2} \right|_0^\pi$$

$$= \frac{1}{r_1 r_2} \left(\sqrt{r_1^2 + r_2^2 + 2r_1 r_2} - \sqrt{r_1^2 + r_2^2 - 2r_1 r_2} \right)$$

$$= \frac{1}{r_1 r_2} \left[(r_1 + r_2) - |r_1 - r_2| \right] = \begin{cases} 2/r_1, & r_2 < r_1, \\ 2/r_2, & r_2 > r_1. \end{cases} \tag{8.24}$$

Thus

$$I_2 = 4\pi \left(\frac{1}{r_1} \int_0^{r_1} e^{-4r_2/a} r_2^2 \, dr_2 + \int_{r_1}^\infty e^{-4r_2/a} r_2 \, dr_2 \right)$$

$$= \frac{\pi a^3}{8 r_1} \left[1 - \left(1 + \frac{2r_1}{a} \right) e^{-4r_1/a} \right]. \tag{8.25}$$

It follows that $\langle V_{ee} \rangle$ is equal to

$$\left(\frac{e^2}{4\pi\epsilon_0} \right) \left(\frac{8}{\pi a^3} \right) \int \left[1 - \left(1 + \frac{2r_1}{a} \right) e^{-4r_1/a} \right] e^{-4r_1/a} r_1 \sin\theta_1 \, dr_1 d\theta_1 d\phi_1.$$

The angular integrals are easy (4π), and the r_1 integral becomes

$$\int_0^\infty \left[r e^{-4r/a} - \left(r + \frac{2r^2}{a} \right) e^{-8r/a} \right] dr = \frac{5a^2}{128}.$$

Finally, then,

$$\langle V_{ee} \rangle = \frac{5}{4a}\left(\frac{e^2}{4\pi\epsilon_0}\right) = -\frac{5}{2}E_1 = 34\ \text{eV}, \tag{8.26}$$

and therefore

$$\langle H \rangle = -109\ \text{eV} + 34\ \text{eV} = -75\ \text{eV}. \tag{8.27}$$

Not bad (remember, the experimental value is -79 eV). But we can do better.

We need to think up a more realistic trial function than ψ_0 (which treats the two electrons as though they did not interact at all). Rather than completely *ignoring* the influence of the other electron, let us say that, on the average, each electron represents a cloud of negative charge which partially *shields* the nucleus, so that the other electron actually sees an *effective* nuclear charge (Z) that is somewhat *less* than 2. This suggests that we use a trial function of the form

$$\psi_1(\mathbf{r}_1, \mathbf{r}_2) \equiv \frac{Z^3}{\pi a^3} e^{-Z(r_1+r_2)/a}. \tag{8.28}$$

We'll treat Z as a variational parameter, picking the value that minimizes $\langle H \rangle$. (Please note that in the variational method we *never touch the Hamiltonian itself*—the Hamiltonian for helium is, and remains, Equation 8.15. But it's fine to *think* about approximating the Hamiltonian *as a way of motivating the choice of the trial wave function.*)

This wave function is an eigenstate of the "unperturbed" Hamiltonian (neglecting electron repulsion), only with Z, instead of 2, in the Coulomb terms. With this in mind, we rewrite H (Equation 8.15) as follows:

$$\begin{aligned} H = &-\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) - \frac{e^2}{4\pi\epsilon_0}\left(\frac{Z}{r_1} + \frac{Z}{r_2}\right) \\ &+ \frac{e^2}{4\pi\epsilon_0}\left(\frac{(Z-2)}{r_1} + \frac{(Z-2)}{r_2} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}\right). \end{aligned} \tag{8.29}$$

The expectation value of H is evidently

$$\langle H \rangle = 2Z^2 E_1 + 2\,(Z-2)\left(\frac{e^2}{4\pi\epsilon_0}\right)\left\langle\frac{1}{r}\right\rangle + \langle V_{ee}\rangle. \tag{8.30}$$

Here $\langle 1/r \rangle$ is the expectation value of $1/r$ in the (one-particle) hydrogenic ground state ψ_{100} (with nuclear charge Z); according to Equation 7.56,

$$\left\langle\frac{1}{r}\right\rangle = \frac{Z}{a}. \tag{8.31}$$

The expectation value of V_{ee} is the same as before (Equation 8.26), except that instead of $Z = 2$ we now want *arbitrary* Z—so we multiply a by $2/Z$:

$$\langle V_{ee} \rangle = \frac{5Z}{8a}\left(\frac{e^2}{4\pi\epsilon_0}\right) = -\frac{5Z}{4}E_1. \tag{8.32}$$

Putting all this together, we find

$$\langle H \rangle = \left[2Z^2 - 4Z\,(Z-2) - (5/4)\,Z\right]E_1 = \left[-2Z^2 + (27/4)\,Z\right]E_1. \tag{8.33}$$

According to the variational principle, this quantity exceeds E_{gs} for *any* value of Z. The *lowest* upper bound occurs when $\langle H \rangle$ is minimized:

$$\frac{d}{dZ}\langle H \rangle = \left[-4Z + (27/4)\right] E_1 = 0,$$

from which it follows that

$$Z = \frac{27}{16} = 1.69. \tag{8.34}$$

This seems reasonable; it tells us that the other electron partially screens the nucleus, reducing its effective charge from 2 down to about 1.69. Putting in this value for Z, we find

$$\langle H \rangle = \frac{1}{2}\left(\frac{3}{2}\right)^6 E_1 = -77.5 \text{ eV}. \tag{8.35}$$

The ground state of helium has been calculated with great precision in this way, using increasingly complicated trial wave functions, with more and more adjustable parameters.[10] But we're within 2% of the correct answer, and, frankly, at this point my own interest in the problem begins to wane.[11]

Problem 8.7 Using $E_{\text{gs}} = -79.0$ eV for the ground state energy of helium, calculate the ionization energy (the energy required to remove just *one* electron). *Hint:* First calculate the ground state energy of the helium ion, He^+, with a single electron orbiting the nucleus; then subtract the two energies.

* **Problem 8.8** Apply the techniques of this Section to the H^- and Li^+ ions (each has two electrons, like helium, but nuclear charges $Z = 1$ and $Z = 3$, respectively). Find the effective (partially shielded) nuclear charge, and determine the best upper bound on E_{gs}, for each case. *Comment:* In the case of H^- you should find that $\langle H \rangle > -13.6$ eV, which would appear to indicate that there is no bound state at all, since it would be energetically favorable for one electron to fly off, leaving behind a neutral hydrogen atom. This is not entirely surprising, since the electrons are less strongly attracted to the nucleus than they are in helium, and the electron repulsion tends to break the atom apart. However, it turns out to be incorrect. With a more sophisticated trial wave function (see Problem 8.25) it can be shown that $E_{\text{gs}} < -13.6$ eV, and hence that a bound state *does* exist. It's only *barely* bound, however, and there are no excited bound states,[12] so H^- has no discrete spectrum (all transitions are to and from the continuum). As a result, it is difficult to study in the laboratory, although it exists in great abundance on the surface of the sun.[13]

[10] The classic studies are E. A. Hylleraas, *Z. Phys.* **65**, 209 (1930); C. L. Pekeris, *Phys. Rev.* **115**, 1216 (1959). For more recent work, see footnote 4.

[11] The first excited state of helium can be calculated in much the same way, using a trial wave function orthogonal to the ground state. See Phillip J. E. Peebles, *Quantum Mechanics*, Princeton U.P., Princeton, NJ (1992), Section 40.

[12] Robert N. Hill, *J. Math. Phys.* **18**, 2316 (1977).

[13] For further discussion see Hans A. Bethe and Edwin E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*, Plenum, New York (1977), Section 34.

8.3 THE HYDROGEN MOLECULE ION

Another classic application of the variational principle is to the hydrogen molecule ion, H_2^+, consisting of a single electron in the Coulomb field of two protons (Figure 8.5). I shall assume for the moment that the protons are fixed in position, a specified distance R apart, although one of the most interesting byproducts of the calculation is going to be the actual *value* of R. The Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0}\left(\frac{1}{r} + \frac{1}{r'}\right), \tag{8.36}$$

where r and r' are the distances to the electron from the respective protons. As always, our strategy will be to guess a reasonable trial wave function, and invoke the variational principle to get a bound on the ground state energy. (Actually, our main interest is in finding out whether this system bonds at *all*—that is, whether its energy is less than that of a neutral hydrogen atom plus a free proton. If our trial wave function indicates that there *is* a bound state, a *better* trial function can only make the bonding even stronger.)

To construct the trial wave function, imagine that the ion is formed by taking a hydrogen atom in its ground state (Equation 4.80),

$$\psi_0(\mathbf{r}) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}, \tag{8.37}$$

bringing the second proton in from "infinity," and nailing it down a distance R away. If R is substantially greater than the Bohr radius, the electron's wave function probably isn't changed very much. But we would like to treat the two protons on an equal footing, so that the electron has the same probability of being associated with either one. This suggests that we consider a trial function of the form

$$\psi = A\left[\psi_0(r) + \psi_0(r')\right]. \tag{8.38}$$

(Quantum chemists call this the **LCAO** technique, because we are expressing the *molecular* wave function as a **l**inear **c**ombination of **a**tomic **o**rbitals.)

Our first task is to *normalize* the trial function:

$$\begin{aligned} 1 = \int |\psi|^2\, d^3\mathbf{r} = |A|^2 \bigg[\int \psi_0(r)^2\, d^3\mathbf{r} \\ + \int \psi_0(r')^2\, d^3\mathbf{r} + 2\int \psi_0(r)\psi_0(r')\, d^3\mathbf{r}\bigg]. \end{aligned} \tag{8.39}$$

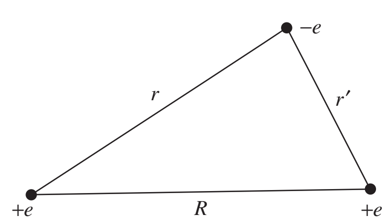

Figure 8.5: The hydrogen molecule ion, H_2^+.

The first two integrals are 1 (since ψ_0 itself is normalized); the third is more tricky. Let

$$I \equiv \langle \psi_0(r) \mid \psi_0(r') \rangle = \frac{1}{\pi a^3} \int e^{-(r+r')/a} \, d^3\mathbf{r}. \tag{8.40}$$

Picking coordinates so that proton 1 is at the origin and proton 2 is on the z axis at the point R (Figure 8.6), we have

$$r' = \sqrt{r^2 + R^2 - 2rR\cos\theta}, \tag{8.41}$$

and therefore

$$I = \frac{1}{\pi a^3} \int e^{-r/a} e^{-\sqrt{r^2+R^2-2rR\cos\theta}/a} r^2 \sin\theta \, dr \, d\theta \, d\phi. \tag{8.42}$$

The ϕ integral is trivial (2π). To do the θ integral, let $y \equiv \sqrt{r^2 + R^2 - 2rR\cos\theta}$, so that $d(y^2) = 2y\,dy = 2rR\sin\theta\,d\theta$. Then

$$\int_0^\pi e^{-\sqrt{r^2+R^2-2rR\cos\theta}/a} \sin\theta \, d\theta = \frac{1}{rR} \int_{|r-R|}^{r+R} e^{-y/a} y \, dy$$

$$= -\frac{a}{rR} \left[e^{-(r+R)/a} (r + R + a) - e^{-|r-R|/a} (|r - R| + a) \right].$$

The r integral is now straightforward:

$$I = \frac{2}{a^2 R} \left[-e^{-R/a} \int_0^\infty (r + R + a) e^{-2r/a} r \, dr + e^{-R/a} \int_0^R (R - r + a) \, r \, dr \right.$$
$$\left. + e^{R/a} \int_R^\infty (r - R + a) \, e^{-2r/a} r \, dr \right].$$

Evaluating the integrals, we find (after some algebraic simplification),

$$I = e^{-R/a} \left[1 + \left(\frac{R}{a}\right) + \frac{1}{3}\left(\frac{R}{a}\right)^2 \right]. \tag{8.43}$$

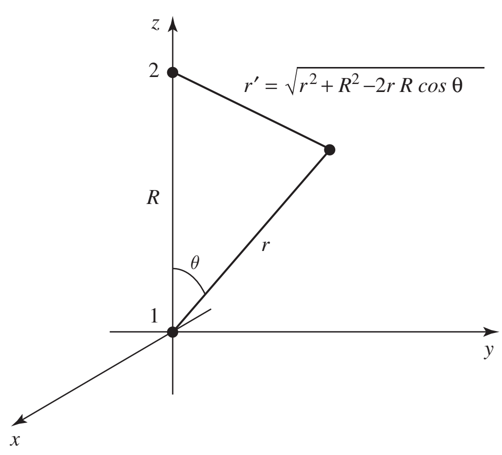

Figure 8.6: Coordinates for the calculation of I (Equation 8.40).

I is called an **overlap** integral; it measures the amount by which $\psi_0(r)$ overlaps $\psi_0(r')$ (notice that it goes to 1 as $R \to 0$, and to 0 as $R \to \infty$). In terms of I, the normalization factor (Equation 8.39) is

$$|A|^2 = \frac{1}{2(1+I)}. \tag{8.44}$$

Next we must calculate the expectation value of H in the trial state ψ. Noting that

$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0}\frac{1}{r}\right)\psi_0(r) = E_1\psi_0(r)$$

(where $E_1 = -13.6$ eV is the ground state energy of atomic hydrogen)—and the same with r' in place of r—we have

$$\begin{aligned} H\psi &= A\left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0}\left(\frac{1}{r}+\frac{1}{r'}\right)\right]\left[\psi_0(r)+\psi_0(r')\right] \\ &= E_1\psi - A\left(\frac{e^2}{4\pi\epsilon_0}\right)\left[\frac{1}{r'}\psi_0(r) + \frac{1}{r}\psi_0(r')\right]. \end{aligned}$$

It follows that

$$\langle H\rangle = E_1 - 2|A|^2\left(\frac{e^2}{4\pi\epsilon_0}\right)\left[\left\langle\psi_0(r)\left|\frac{1}{r'}\right|\psi_0(r)\right\rangle + \left\langle\psi_0(r)\left|\frac{1}{r}\right|\psi_0(r')\right\rangle\right]. \tag{8.45}$$

I'll let you calculate the two remaining quantities, the so-called **direct integral**,

$$D \equiv a\left\langle\psi_0(r)\left|\frac{1}{r'}\right|\psi_0(r)\right\rangle, \tag{8.46}$$

and the **exchange integral**,

$$X \equiv a\left\langle\psi_0(r)\left|\frac{1}{r}\right|\psi_0(r')\right\rangle. \tag{8.47}$$

The results (see Problem 8.9) are

$$D = \frac{a}{R} - \left(1+\frac{a}{R}\right)e^{-2R/a}, \tag{8.48}$$

and

$$X = \left(1+\frac{R}{a}\right)e^{-R/a}. \tag{8.49}$$

Putting all this together, and recalling (Equations 4.70 and 4.72) that $E_1 = -\left(e^2/4\pi\epsilon_0\right)(1/2a)$, we conclude:

$$\langle H\rangle = \left[1 + 2\frac{(D+X)}{(1+I)}\right]E_1. \tag{8.50}$$

According to the variational principle, the ground state energy is *less* than $\langle H\rangle$. Of course, this is only the *electron's* energy—there is also potential energy associated with the proton–proton repulsion:

$$V_{pp} = \frac{e^2}{4\pi\epsilon_0}\frac{1}{R} = -\frac{2a}{R}E_1. \tag{8.51}$$

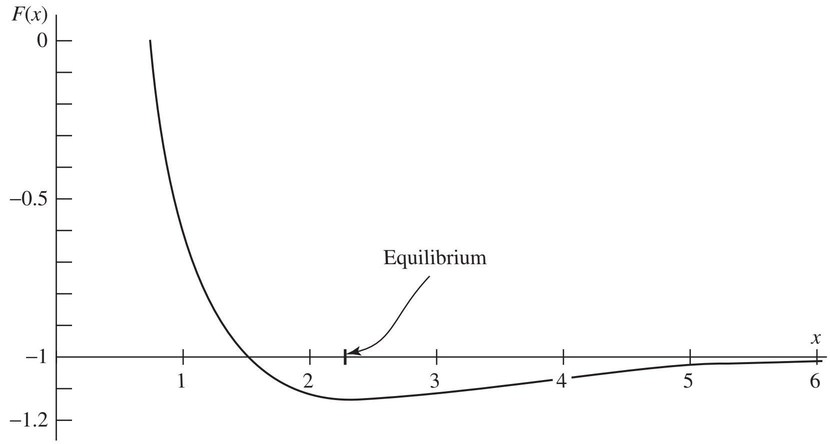

Figure 8.7: Plot of the function $F(x)$, Equation 8.52, showing existence of a bound state.

Thus the *total* energy of the system, in units of $-E_1$, and expressed as a function of $x \equiv R/a$, is less than

$$F(x) = -1 + \frac{2}{x}\left\{\frac{\left(1-(2/3)x^2\right)e^{-x} + (1+x)\,e^{-2x}}{1+\left(1+x+(1/3)x^2\right)e^{-x}}\right\}. \tag{8.52}$$

This function is plotted in Figure 8.7. Evidently bonding *does* occur, for there exists a region in which the graph goes below -1, indicating that the energy is less than that of a neutral atom plus a free proton (-13.6 eV). It's a **covalent bond**, with the electron shared equally by the two protons. The equilibrium separation of the protons is about 2.4 Bohr radii, or 1.3 Å (the experimental value is 1.06 Å). The calculated binding energy is 1.8 eV, whereas the experimental value is 2.8 eV (the variational principle, as always, *over* estimates the ground state energy—and hence *under* estimates the strength of the bond—but never mind: The essential point was to see whether binding occurs at all; a better variational function can only make the potential well even deeper.

* **Problem 8.9** Evaluate D and X (Equations 8.46 and 8.47). Check your answers against Equations 8.48 and 8.49.

** **Problem 8.10** Suppose we used a *minus* sign in our trial wave function (Equation 8.38):

$$\psi = A\left[\psi_0(r) - \psi_0\left(r'\right)\right]. \tag{8.53}$$

Without doing any new integrals, find $F(x)$ (the analog to Equation 8.52) for this case, and construct the graph. Show that there is no evidence of bonding.[14] (Since the variational principle only gives an *upper bound*, this doesn't *prove* that bonding cannot occur for such a state, but it certainly doesn't look promising.)

[14] The wave function with the plus sign (Equation 8.38) is called the **bonding orbital**. Bonding is associated with a buildup of electron probability in between the two nuclei. The odd linear combination (Equation 8.53) has a *node* at the center, so it's not surprising that this configuration doesn't lead to bonding; it is called the **anti-bonding orbital**.

*** **Problem 8.11** The second derivative of $F(x)$, at the equilibrium point, can be used to estimate the natural frequency of vibration (ω) of the two protons in the hydrogen molecule ion (see Section 2.3). If the ground state energy ($\hbar\omega/2$) of this oscillator exceeds the binding energy of the system, it will fly apart. Show that in fact the oscillator energy is small enough that this will *not* happen, and estimate how many bound vibrational levels there are. *Note*: You're not going to be able to obtain the position of the minimum—still less the second derivative at that point—analytically. Do it numerically, on a computer.

8.4 THE HYDROGEN MOLECULE

Now consider the hydrogen molecule itself, adding a second electron to the hydrogen molecule ion we studied in Section 8.3. Taking the two protons to be at rest, the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) + \frac{e^2}{4\pi\epsilon_0}\left(\frac{1}{r_{12}} + \frac{1}{R} - \frac{1}{r_1} - \frac{1}{r_1'} - \frac{1}{r_2} - \frac{1}{r_2'}\right) \tag{8.54}$$

where r_1 and r_1' are the distances of electron 1 from each proton and r_2 and r_2' are the distances of electron 2 from each proton; as shown in Figure 8.8. The six potential energy terms describe the repulsion between the two electrons, the repulsion between the two protons, and the attraction of each electron to each proton.

For the variational wave function, associate one electron with each proton, and symmetrize:

$$\psi_+(\mathbf{r}_1, \mathbf{r}_2) = A_+ \left[\psi_0(r_1)\ \psi_0\left(r_2'\right) + \psi_0\left(r_1'\right)\ \psi_0(r_2)\right]. \tag{8.55}$$

We'll calculate the normalization A_+ in a moment. Since this spatial wave function is symmetric under interchange, the electrons must occupy the antisymmetric (singlet) spin state. Of course, we could also choose the trial wave function

$$\psi_-(\mathbf{r}_1, \mathbf{r}_2) = A_- \left[\psi_0(r_1)\ \psi_0\left(r_2'\right) - \psi_0\left(r_1'\right)\ \psi_0(r_2)\right] \tag{8.56}$$

in which case the electrons would be in a symmetric (triplet) spin state. These two variational wave functions constitute the **Heitler–London approximation**.[15] It is not obvious which of

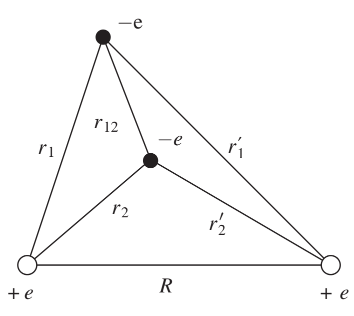

Figure 8.8: Diagram of H_2 showing the distances on which the potential energy depends.

[15] W. Heitler and F. London, *Z. Phys.* **44**, 455 (1928). For an English translation see Hinne Hettema, *Quantum Chemistry: Classic Scientific Papers*, World Scientific, New Jersey, PA, 2000.

Equations 8.55 or 8.56 would be energetically favored, so let's calculate the energy of each one, and find out.[16]

First we need to normalize the wave functions. Note that

$$|\psi_\pm(\mathbf{r}_1, \mathbf{r}_2)|^2 = A_\pm^2 \left[\psi_0(r_1)^2 \, \psi_0\left(r_2'\right)^2 + \psi_0\left(r_1'\right)^2 \psi_0(r_2)^2 \pm 2\, \psi_0(r_1)\, \psi_0\left(r_2'\right) \psi_0\left(r_1'\right) \psi_0(r_2)\right]. \tag{8.58}$$

Normalization requires

$$\begin{aligned} 1 &= \int\int |\psi_\pm(\mathbf{r}_1, \mathbf{r}_2)|^2 \, d^3\mathbf{r}_1 \, d^3\mathbf{r}_2 \\ &= A_\pm^2 \left[\int \psi_0(r_1)^2 \, d^3\mathbf{r}_1 \int \psi_0\left(r_2'\right)^2 d^3\mathbf{r}_2 + \int \psi_0\left(r_1'\right)^2 d^3\mathbf{r}_1 \int \psi_0(r_2)^2 \, d^3\mathbf{r}_2 \right. \\ &\quad \left. \pm\, 2 \int \psi_0(r_1)\, \psi_0\left(r_1'\right) d^3\mathbf{r}_1 \int \psi_0\left(r_2'\right) \psi_0(r_2)\, d^3\mathbf{r}_2 \right]. \end{aligned} \tag{8.59}$$

The individual orbitals are normalized and the overlap integral was given the symbol I and calculated in Equation 8.43. Thus

$$A_\pm = \frac{1}{\sqrt{2\left(1 \pm I^2\right)}}. \tag{8.60}$$

To calculate the expectation value of the energy, we will start with the kinetic energy of particle 1. Since ψ_0 is the ground state of the hydrogen Hamiltonian, the same trick that brought us to Equation 8.45 gives

$$\begin{aligned} -\frac{\hbar^2}{2m} \nabla_1^2 \psi_\pm &= A_\pm \left[\left(-\frac{\hbar^2}{2m} \nabla_1^2 \psi_0(r_1)\right) \psi_0\left(r_2'\right) \pm \left(-\frac{\hbar^2}{2m} \nabla_1^2 \psi_0\left(r_1'\right)\right) \psi_0(r_2)\right] \\ &= A_\pm \left[\left(E_1 + \frac{e^2}{4\pi \epsilon_0 r_1}\right) \psi_0(r_1)\, \psi_0\left(r_2'\right) \right. \\ &\quad \left. \pm \left(E_1 + \frac{e^2}{4\pi \epsilon_0 r_1'}\right) \psi_0\left(r_1'\right) \psi_0(r_2)\right] \\ &= E_1 \psi_\pm + \frac{e^2}{4\pi \epsilon_0 a} A_\pm \left(\frac{a}{r_1} \psi_0(r_1)\, \psi_0\left(r_2'\right) \pm \frac{a}{r_1'} \psi_0\left(r_1'\right) \psi_0(r_2)\right). \end{aligned}$$

Taking the inner product with $\psi_\pm$ then gives

$$\begin{aligned} \left\langle -\frac{\hbar^2}{2m} \nabla_1^2 \right\rangle = E_1 + \left(\frac{e^2}{4\pi \epsilon_0 a}\right) A_\pm^2 &\left[\left\langle \psi_0(r_1) \left| \frac{a}{r_1} \right| \psi_0(r_1) \right\rangle \left\langle \psi_0\left(r_2'\right) \middle| \psi_0\left(r_2'\right)\right\rangle \right. \\ &\pm \left\langle \psi_0\left(r_1'\right) \left| \frac{a}{r_1} \right| \psi_0(r_1) \right\rangle \left\langle \psi_0(r_2) \middle| \psi_0\left(r_2'\right)\right\rangle \end{aligned}$$

---

[16] Another natural variational wave function consists of placing both electrons in the bonding orbital studied in Section 8.3:

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \left(\frac{\psi_0(r_1) + \psi_0\left(r_1'\right)}{\sqrt{2(1+I)}}\right) \left(\frac{\psi_0(r_2) + \psi_0\left(r_2'\right)}{\sqrt{2(1+I)}}\right), \tag{8.57}$$

also paired with a singlet spin state. If you expand this function you'll see that half the terms—such as $\psi_0(r_1)\, \psi_0(r_2)$—involve attaching two electrons to the same proton, which is energetically costly because of the electron–electron repulsion in Equation 8.54. The Heitler–London approximation, Equation 8.55, amounts to dropping the offending terms from Equation 8.57.

$$\pm\left\langle\psi_0(r_1)\left|\frac{a}{r_1'}\right|\psi_0\left(r_1'\right)\right\rangle\left\langle\psi_0\left(r_2'\right)\mid\psi_0(r_2)\right\rangle$$
$$+\left\langle\psi_0\left(r_1'\right)\left|\frac{a}{r_1'}\right|\psi_0\left(r_1'\right)\right\rangle\left\langle\psi_0(r_2)\mid\psi_0(r_2)\right\rangle\Bigg]. \tag{8.61}$$

These inner products were calculated in Section 8.3 and the kinetic energy of particle 1 is

$$\left\langle-\frac{\hbar^2}{2m}\nabla_1^2\right\rangle=E_1+\left(\frac{e^2}{4\pi\epsilon_0 a}\right)\frac{1\pm I\,X}{1\pm I^2}. \tag{8.62}$$

The kinetic energy of particle 2 is of course the same, so the total kinetic energy is simply twice Equation 8.62. The calculation of the electron–proton potential energy is similar; you will show in Problem 8.13 that

$$\left\langle-\frac{e^2}{4\pi\epsilon_0 r_1}\right\rangle=-\frac{1}{2}\left(\frac{e^2}{4\pi\epsilon_0 a}\right)\frac{1+D\pm 2\,I\,X}{1\pm I^2} \tag{8.63}$$

and the total electron–proton potential energy is four times this amount.

The electron–electron potential energy is given by

$$\begin{aligned}\langle V_{ee}\rangle &=\left(\frac{e^2}{4\pi\epsilon_0 a}\right)\int\int|\psi_\pm(\mathbf{r}_1,\mathbf{r}_2)|^2\,\frac{a}{r_{12}}\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2\\ &=\left(\frac{e^2}{4\pi\epsilon_0 a}\right)A_\pm^2\left[\int\int\psi_0(r_1)^2\,\frac{a}{r_{12}}\,\psi_0\left(r_2'\right)^2\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2\right.\\ &+\int\int\psi_0\left(r_1'\right)^2\,\frac{a}{r_{12}}\,\psi_0(r_2)^2\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2\\ &\left.\pm 2\int\int\psi_0(r_1)\,\psi_0\left(r_1'\right)\,\frac{a}{r_{12}}\,\psi_0(r_2)\,\psi_0\left(r_2'\right)\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2\right].\end{aligned} \tag{8.64}$$

The first two integrals in Equation 8.64 are equal, as you can see by interchanging the labels 1 and 2. We will give the two remaining integrals the names

$$D_2=\int\int|\psi_0(r_1)|^2\,\frac{a}{r_{12}}\,\left|\psi_0\left(r_2'\right)\right|^2\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2 \tag{8.65}$$

$$X_2=\int\int\psi_0(r_1)\,\psi_0\left(r_1'\right)\,\frac{a}{r_{12}}\,\psi_0(r_2)\,\psi_0\left(r_2'\right)\,d^3\mathbf{r}_1\,d^3\mathbf{r}_2 \tag{8.66}$$

so that

$$\langle V_{ee}\rangle=\left(\frac{e^2}{4\pi\epsilon_0 a}\right)\frac{D_2\pm X_2}{1\pm I^2}. \tag{8.67}$$

The evaluation of these integrals is discussed in Problem 8.14. Note that the integral D_2 is just the electrostatic potential energy of two charge distributions $\rho_1=|\psi_0(r_1)|^2$ and $\rho_2=|\psi_0(r_2)|^2$. The exchange term X_2 has no such classical counterpart.

When we add all of the contributions to the energy—the kinetic energy, the electron–proton potential energy, the electron–electron potential energy, and the proton–proton potential energy (which is a constant, $e^2/4\pi\epsilon_0 R$)—we get

$$\boxed{\langle H\rangle_{\pm} = 2\,E_1\left[1 - \frac{a}{R} + \frac{2\,D - D_2 \pm (2\,I\,X - X_2)}{1 \pm I^2}\right].} \tag{8.68}$$

A plot of $\langle H\rangle_+$ and $\langle H\rangle_-$ is shown in Figure 8.9. Recall that the state ψ_+ requires placing the two electrons in the singlet spin configuration, whereas ψ_- means putting them in a triplet spin configuration. According to the figure, bonding only occurs if the two electrons are in a singlet configuration—something that is confirmed experimentally. Again, it's a covalent bond.

Locating the minimum on the plot, our calculation predicts a bond length of 1.64 Bohr radii (the experimental value is 1.40 Bohr radii), and suggests a binding energy of 3.15 eV (whereas the experimental value is 4.75 eV). The trends here follow those of the Hydrogen molecule ion: the calculation overestimates the bond length and underestimates the binding energy, but the agreement is surprisingly good for a variational calculation with no adjustable parameters.

The difference between the singlet and triplet energies is called the **exchange splitting** J. In the Heitler–London approximation it is

$$J = \langle H\rangle_+ - \langle H\rangle_- = 4\,E_1\,\frac{(D_2 - 2\,D)\;I^2 - (X_2 - 2\,I\,X)}{1 - I^4}, \tag{8.69}$$

which is roughly $-10\,\text{eV}$ (negative because the singlet is lower in energy) at the equilibrium separation. This means a strong preference for having the electron spins anti-aligned. But in this treatment of H_2 we've left out completely the (magnetic) spin–spin interaction between the electrons—remember that the spin–spin interaction between the *proton* and the electron is what leads to hyperfine splitting (Section 7.5). Were we right to ignore it here? Absolutely: applying Equation 7.92 to two electrons a distance R apart, the energy of the spin–spin interaction is something like 10^{-4} eV in this system, five orders of magnitude smaller than the exchange splitting.

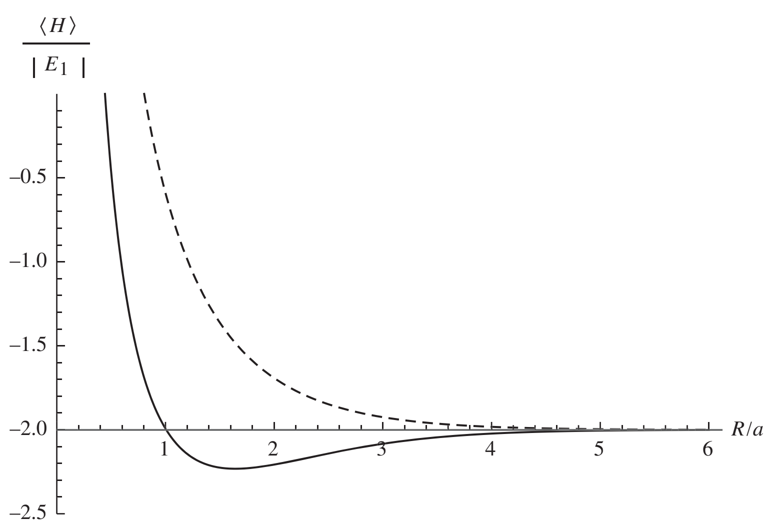

Figure 8.9: The total energy of the singlet (solid curve) and triplet (dashed curve) states for H_2, as a function of the separation R between the protons. The singlet state has a minimum at around 1.6 Bohr radii, representing a stable bond. The triplet state is unstable and will dissociate, as the energy is minimized for $R \to \infty$.

This calculation shows us that different spin configurations can have very different energies, even when the interaction between the spins is negligible. And that helps us understand ferromagnetism (where the spins in a material align) and anti-ferromagnetism (where the spins alternate). As we've just seen, the spin–spin interaction is *way* too weak to account for this—but the exchange splitting isn't. Counterintuitively, it's not a *magnetic* interaction that accounts for ferromagnetism, but an electrostatic one! H_2 is a sort of inchoate anti-ferromagnet where the Hamiltonian, which is independent of the spin, selects a certain *spatial* ground state and the *spin* state comes along for the ride, to satisfy the Fermi statistics.

Problem 8.12 Show that the antisymmetric state (Equation 8.56) can be expressed in terms of the molecular orbitals of Section 8.3—specifically, by placing one electron in the bonding orbital (Equation 8.38) and one in the anti-bonding orbital (Equation 8.53).

Problem 8.13 Verify Equation 8.63 for the electron–proton potential energy.

∗∗∗ **Problem 8.14** The two-body integrals D_2 and X_2 are defined in Equations 8.65 and 8.66. To evaluate D_2 we write

$$D_2 = \int \left|\psi_0\left(r_2'\right)\right|^2 \Phi(r_2)\, d^3\mathbf{r}_2$$
$$= \int\int\int \frac{e^{-2\sqrt{R^2+r_2^2-2\,R\,r_2\,\cos\theta_2}/a}}{\pi\, a^3}\, \Phi(r_2)\, r_2^2\, dr_2\, \sin\theta_2\, d\theta_2\, d\phi_2$$

where θ_2 is the angle between $\boldsymbol{R}$ and $\boldsymbol{r_2}$ (Figure 8.8), and

$$\Phi(r_2) \equiv \int |\psi_0(r_1)|^2 \frac{a}{|\mathbf{r}_1 - \mathbf{r}_2|}\, d^3\mathbf{r}_1 \,.$$

(a) Consider first the integral over $\mathbf{r}_1$. Align the z axis with $\mathbf{r}_2$ (which is a constant vector for the purposes of this first integral) so that

$$\Phi(r_2) = \frac{1}{\pi\, a^3}\int\int\int \frac{a e^{-2\,r_1/a}}{\sqrt{r_1^2 + r_2^2 - 2\,r_1\,r_2\,\cos\theta_1}}\, r_1^2\, dr_1\, \sin\theta_1\, d\theta_1\, d\phi_1 \,.$$

Do the angular integration first and show that

$$\Phi(r_2) = \frac{a}{r_2} - \left(1 + \frac{a}{r_2}\right) e^{-2\,r_2/a} \,.$$

(b) Plug your result from part (a) back into the relation for D_2, and show that

$$D_2 = \frac{a}{R} - e^{-2\,R/a}\left[\frac{1}{6}\left(\frac{R}{a}\right)^2 + \frac{3}{4}\left(\frac{R}{a}\right) + \frac{11}{8} + \frac{a}{R}\right]. \qquad (8.70)$$

Again, do the angular integration first.

Comment: The integral X_2 *can* also be evaluated in closed form, but the procedure is rather involved.[17] We will simply quote the result,

[17] The calculation was done by Y. Sugiura, *Z. Phys.* **44**, 455 (1927).

$$X_2 = e^{-2R/a}\left[\frac{5}{8} - \frac{23}{20}\frac{R}{a} - \frac{3}{5}\left(\frac{R}{a}\right)^2 - \frac{1}{15}\left(\frac{R}{a}\right)^3\right]$$
$$+\frac{6}{5}\frac{a}{R}I^2\left[\gamma + \log\left(\frac{R}{a}\right) + \left(\frac{\tilde{I}}{I}\right)^2 \mathrm{Ei}\left(-\frac{4R}{a}\right) - 2\frac{\tilde{I}}{I}\mathrm{Ei}\left(-\frac{2R}{a}\right)\right], \tag{8.71}$$

where $\gamma = 0.5772\ldots$ is Euler's constant, $\mathrm{Ei}(x)$ is the exponential integral

$$\mathrm{Ei}(x) = -\int_{-x}^{\infty} \frac{e^{-t}}{t}\,dt\,,$$

and $\tilde{I}$ is obtained from I by switching the sign of R:

$$\tilde{I} = e^{R/a}\left[1 - \frac{R}{a} + \frac{1}{3}\left(\frac{R}{a}\right)^2\right]. \tag{8.72}$$

* **Problem 8.15** Make a plot of the kinetic energy for both the singlet and triplet states of H_2, as a function of R/a. Do the same for the electron-proton potential energy and for the electron–electron potential energy. You should find that the triplet state has lower potential energy than the singlet state for all values of R. However, the singlet state's *kinetic* energy is *so* much smaller that its total energy comes out lower. *Comment*: In situations where there is not a large kinetic energy cost to aligning the spins, such as two electrons in a partially filled orbital in an atom, the triplet state can come out lower in energy. This is the physics behind Hund's first rule.

FURTHER PROBLEMS ON CHAPTER 8

Problem 8.16

(a) Use the function $\psi(x) = Ax(a-x)$ (for $0 < x < a$, otherwise 0) to get an upper bound on the ground state of the infinite square well.

(b) Generalize to a function of the form $\psi(x) = A[x(a-x)]^p$, for some real number p. What is the optimal value of p, and what is the best bound on the ground state energy? Compare the exact value. *Answer:* $\left(5 + 2\sqrt{6}\right)\hbar^2/2ma^2$.

Problem 8.17

(a) Use a trial wave function of the form

$$\psi(x) = \begin{cases} A\cos(\pi x/a), & -a/2 < x < a/2, \\ 0, & \text{otherwise,} \end{cases}$$

to obtain a bound on the ground state energy of the one-dimensional harmonic oscillator. What is the "best" value of a? Compare $\langle H \rangle_{\min}$ with the exact energy. *Note:* This trial function has a "kink" in it (a discontinuous derivative) at $\pm a/2$; do you need to take account of this, as I did in Example 8.3?

(b) Use $\psi(x) = B\sin(\pi x/a)$ on the interval $(-a, a)$ to obtain a bound on the first excited state. Compare the exact answer.

** **Problem 8.18**

(a) Generalize Problem 8.2, using the trial wave function[18]

$$\psi(x) = \frac{A}{\left(x^2 + b^2\right)^n},$$

for arbitrary n. *Partial answer:* The best value of b is given by

$$b^2 = \frac{\hbar}{m\omega}\left[\frac{n(4n-1)(4n-3)}{2(2n+1)}\right]^{1/2}.$$

(b) Find the least upper bound on the first excited state of the harmonic oscillator using a trial function of the form

$$\psi(x) = \frac{Bx}{\left(x^2 + b^2\right)^n}.$$

Partial answer: The best value of b is given by

$$b^2 = \frac{\hbar}{m\omega}\left[\frac{n(4n-5)(4n-3)}{2(2n+1)}\right]^{1/2}.$$

(c) Notice that the bounds approach the exact energies as $n \to \infty$. Why is that? *Hint:* Plot the trial wave functions for $n = 2$, $n = 3$, and $n = 4$, and compare them with the true wave functions (Equations 2.60 and 2.63). To do it analytically, start with the identity

$$e^z = \lim_{n\to\infty}\left(1 + \frac{z}{n}\right)^n.$$

Problem 8.19 Find the lowest bound on the ground state of hydrogen you can get using a gaussian trial wave function

$$\psi(\mathbf{r}) = Ae^{-br^2},$$

where A is determined by normalization and b is an adjustable parameter. *Answer:* -11.5 eV.

Problem 8.20 Find an upper bound on the energy of the first excited state of the hydrogen atom. A trial function with $\ell = 1$ will automatically be orthogonal to the ground state (see footnote 6); for the radial part of ψ you can use the same function as in Problem 8.19.

** **Problem 8.21** If the photon had a nonzero mass $\left(m_\gamma \neq 0\right)$, the Coulomb potential would be replaced by the **Yukawa potential**,

$$V(\mathbf{r}) = -\frac{e^2}{4\pi\epsilon_0}\frac{e^{-\mu r}}{r}, \tag{8.73}$$

where $\mu = m_\gamma c/\hbar$. With a trial wave function of your own devising, estimate the binding energy of a "hydrogen" atom with this potential. Assume $\mu a \ll 1$, and give your answer correct to order $(\mu a)^2$.

[18] W. N. Mei, *Int. J. Educ. Sci. Tech.* **27**, 285 (1996).

Problem 8.22 Suppose you're given a two-level quantum system whose (time-independent) Hamiltonian H^0 admits just two eigenstates, ψ_a (with energy E_a), and ψ_b (with energy E_b). They are orthogonal, normalized, and nondegenerate (assume E_a is the smaller of the two energies). Now we turn on a perturbation H', with the following matrix elements:

$$\langle \psi_a \mid H' \mid \psi_a \rangle = \langle \psi_b \mid H' \mid \psi_b \rangle = 0; \quad \langle \psi_a \mid H' \mid \psi_b \rangle = \langle \psi_b \mid H' \mid \psi_a \rangle = h, \tag{8.74}$$

where h is some specified constant.

(a) Find the exact eigenvalues of the perturbed Hamiltonian.

(b) Estimate the energies of the perturbed system using second-order perturbation theory.

(c) Estimate the ground state energy of the perturbed system using the variational principle, with a trial function of the form

$$\psi = (\cos\phi)\,\psi_a + (\sin\phi)\,\psi_b, \tag{8.75}$$

where ϕ is an adjustable parameter. *Note:* Writing the linear combination in this way is just a neat way to guarantee that ψ is normalized.

(d) Compare your answers to (a), (b), and (c). Why is the variational principle so accurate, in this case?

Problem 8.23 As an explicit example of the method developed in Problem 8.22, consider an electron at rest in a uniform magnetic field $\mathbf{B} = B_z\hat{k}$, for which the Hamiltonian is (Equation 4.158):

$$H^0 = \frac{eB_z}{m} S_z. \tag{8.76}$$

The eigenspinors, χ_a and χ_b, and the corresponding energies, E_a and E_b, are given in Equation 4.161. Now we turn on a perturbation, in the form of a uniform field in the x direction:

$$H' = \frac{eB_x}{m} S_x. \tag{8.77}$$

(a) Find the matrix elements of H', and confirm that they have the structure of Equation 8.74. What is h?

(b) Using your result in Problem 8.22(b), find the new ground state energy, in second-order perturbation theory.

(c) Using your result in Problem 8.22(c), find the variational principle bound on the ground state energy.

∗∗∗ **Problem 8.24** Although the Schrödinger equation for helium itself cannot be solved exactly, there exist "helium-like" systems that do admit exact solutions. A simple example[19] is "rubber-band helium," in which the Coulomb forces are replaced by Hooke's law forces:

[19] For a more sophisticated model, see R. Crandall, R. Whitnell, and R. Bettega, *Am. J. Phys* **52**, 438 (1984).

$$H = -\frac{\hbar^2}{2m}\left(\nabla_1^2 + \nabla_2^2\right) + \frac{1}{2}m\omega^2\left(r_1^2 + r_2^2\right) - \frac{\lambda}{4}m\omega^2|\mathbf{r}_1 - \mathbf{r}_2|^2. \tag{8.78}$$

(a) Show that the change of variables from $\mathbf{r}_1$, $\mathbf{r}_2$, to

$$\mathbf{u} \equiv \frac{1}{\sqrt{2}}(\mathbf{r}_1 + \mathbf{r}_2), \quad \mathbf{v} \equiv \frac{1}{\sqrt{2}}(\mathbf{r}_1 - \mathbf{r}_2), \tag{8.79}$$

turns the Hamiltonian into two independent three-dimensional harmonic oscillators:

$$H = \left[-\frac{\hbar^2}{2m}\nabla_u^2 + \frac{1}{2}m\omega^2 u^2\right] + \left[-\frac{\hbar^2}{2m}\nabla_v^2 + \frac{1}{2}(1-\lambda)\,m\omega^2 v^2\right]. \tag{8.80}$$

(b) What is the *exact* ground state energy for this system?

(c) If we didn't know the exact solution, we might be inclined to apply the method of Section 8.2 to the Hamiltonian in its original form (Equation 8.78). Do so (but don't bother with shielding). How does your result compare with the exact answer? *Answer:* $\langle H\rangle = 3\hbar\omega\,(1 - \lambda/4)$.

∗∗∗ **Problem 8.25** In Problem 8.8 we found that the trial wave function with shielding (Equation 8.28), which worked well for helium, is inadequate to confirm the existence of a bound state for the negative hydrogen ion. Chandrasekhar[20] used a trial wave function of the form

$$\psi(\mathbf{r}_1, \mathbf{r}_2) \equiv A\left[\psi_1(r_1)\psi_2(r_2) + \psi_2(r_1)\psi_1(r_2)\right], \tag{8.81}$$

where

$$\psi_1(r) \equiv \sqrt{\frac{Z_1^3}{\pi a^3}}e^{-Z_1 r/a}, \quad \text{and} \quad \psi_2(r) \equiv \sqrt{\frac{Z_2^3}{\pi a^3}}e^{-Z_2 r/a}. \tag{8.82}$$

In effect, he allowed two *different* shielding factors, suggesting that one electron is relatively close to the nucleus, and the other is farther out. (Because electrons are identical particles, the spatial wave function must be symmetrized with respect to interchange. The *spin* state—which is irrelevant to the calculation—is evidently antisymmetric.) Show that by astute choice of the adjustable parameters Z_1 and Z_2 you can get $\langle H\rangle$ less than -13.6 eV. *Answer:*

$$\langle H\rangle = \frac{E_1}{x^6 + y^6}\left(-x^8 + 2x^7 + \frac{1}{2}x^6y^2 - \frac{1}{2}x^5y^2 - \frac{1}{8}x^3y^4 + \frac{11}{8}xy^6 - \frac{1}{2}y^8\right),$$

where $x \equiv Z_1 + Z_2$ and $y \equiv 2\sqrt{Z_1 Z_2}$. Chandrasekhar used $Z_1 = 1.039$ (since this is larger than 1, the motivating interpretation as an effective nuclear charge cannot be sustained, but never mind—it's still an acceptable trial wave function) and $Z_2 = 0.283$.

Problem 8.26 The fundamental problem in harnessing nuclear fusion is getting the two particles (say, two deuterons) close enough together for the attractive (but short-range) nuclear force to overcome the Coulomb repulsion. The "bulldozer" method is to heat

[20] S. Chandrasekhar, *Astrophys. J.* **100**, 176 (1944).

the particles up to fantastic temperatures, and allow the random collisions to bring them together. A more exotic proposal is **muon catalysis**, in which we construct a "hydrogen molecule ion," only with deuterons in place of protons, and a *muon* in place of the electron. Predict the equilibrium separation distance between the deuterons in such a structure, and explain why muons are superior to electrons for this purpose.[21]

∗∗∗ **Problem 8.27 Quantum dots.** Consider a particle constrained to move in two dimensions in the cross-shaped region shown in Figure 8.10. The "arms" of the cross continue out to infinity. The potential is zero within the cross, and infinite in the shaded areas outside. Surprisingly, this configuration admits a positive-energy bound state.[22]

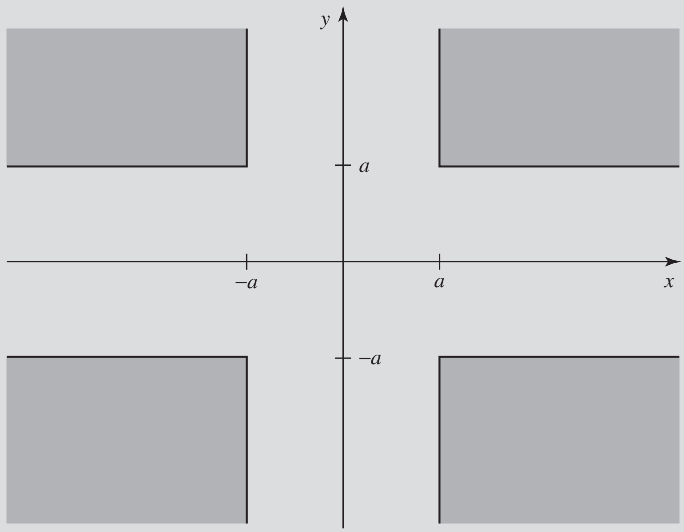

Figure 8.10: The cross-shaped region for Problem 8.27.

(a) Show that the lowest energy that can propagate off to infinity is

$$E_{\text{threshold}} = \frac{\pi^2\hbar^2}{8ma^2};$$

any solution with energy *less* than that has to be a bound state. *Hint:* Go way out one arm (say, $x \gg a$), and solve the Schrödinger equation by separation of variables; if the wave function propagates out to infinity, the dependence on x must take the form $\exp(ik_x x)$ with $k_x > 0$.

[21] The classic paper on muon-catalyzed fusion is J. D. Jackson, *Phys. Rev.* **106**, 330 (1957); for a more recent popular review, see J. Rafelski and S. Jones, *Scientific American*, November 1987, page 84.

[22] This model is taken from R. L. Schult *et al.*, *Phys. Rev. B* **39**, 5476 (1989). For further discussion see J. T. Londergan and D. P. Murdock, *Am. J. Phys.* **80**, 1085 (2012). In the presence of quantum tunneling a classically bound state can become unbound; this is the reverse: A classically *un*bound state is quantum mechanically *bound*.

(b) Now use the variational principle to show that the ground state has energy less than $E_{\text{threshold}}$. Use the following trial wave function (suggested by Jim McTavish):

$$\psi(x,y) = A\begin{cases}\left[\cos(\pi x/2a)+\cos(\pi y/2a)\right]e^{-\alpha}, & |x|\le a \text{ and } |y|\le a\\ \cos(\pi x/2a)e^{-\alpha|y|/a}, & |x|\le a \text{ and } |y|> a\\ \cos(\pi y/2a)e^{-\alpha|x|/a}, & |x|> a \text{ and } |y|\le a\\ 0, & \text{elsewhere.}\end{cases}$$

Normalize it to determine A, and calculate the expectation value of H. *Answer:*

$$\langle H\rangle = \frac{\hbar^2}{ma^2}\left[\frac{\pi^2}{8}-\left(\frac{1-(\alpha/4)}{1+\left(8/\pi^2\right)+(1/2\alpha)}\right)\right].$$

Now minimize with respect to α, and show that the result is less than $E_{\text{threshold}}$. *Hint:* Take full advantage of the symmetry of the problem—you only need to integrate over 1/8 of the open region, since the other seven integrals will be the same. Note however that whereas the trial wave function is continuous, its *derivatives* are *not*—there are "roof-lines" at the joins, and you will need to exploit the technique of Example 8.3.[23]

Problem 8.28 In Yukawa's original theory (1934), which remains a useful approximation in nuclear physics, the "strong" force between protons and neutrons is mediated by the exchange of π-mesons. The potential energy is

$$V(r) = -r_0 V_0 \frac{e^{-r/r_0}}{r}, \tag{8.83}$$

where r is the distance between the nucleons, and the range r_0 is related to the mass of the meson: $r_0 = \hbar/m_\pi c$. *Question:* Does this theory account for the existence of the **deuteron** (a bound state of the proton and the neutron)?

The Schrödinger equation for the proton/neutron system is (see Problem 5.1):

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi(\mathbf{r}) + V(r)\psi(\mathbf{r}) = E\psi(\mathbf{r}), \tag{8.84}$$

where μ is the reduced mass (the proton and neutron have almost identical masses, so call them both m), and $\mathbf{r}$ is the position of the neutron (say) relative to the proton: $\mathbf{r} = \mathbf{r}_n - \mathbf{r}_p$. Your task is to show that there exists a solution with negative energy (a bound state), using a variational trial wave function of the form

$$\psi_\beta(\mathbf{r}) = Ae^{-\beta r/r_0}. \tag{8.85}$$

[23] W.-N. Mei gets a somewhat better bound (and avoids the roof-lines) using

$$\psi(x,y) = Ae^{-\alpha\left(x^2+y^2\right)/a^2}\begin{cases}\left(1-x^2y^2/a^4\right),\\ \left(1-x^2/a^2\right),\\ \left(1-y^2/a^2\right),\end{cases}$$

but the integrals have to be done numerically.

(a) Determine A, by normalizing $\psi_\beta(\mathbf{r})$.

(b) Find the expectation value of the Hamiltonian $\left(H = -\frac{\hbar^2}{2\mu}\nabla^2 + V\right)$ in the state ψ_β. *Answer:*

$$E(\beta) = \frac{\hbar^2}{2\mu r_0^2}\beta^2\left[1 - \frac{4\gamma\beta}{(1+2\beta)^2}\right], \quad \text{where } \gamma \equiv \frac{2\mu r_0^2}{\hbar^2}V_0. \tag{8.86}$$

(c) Optimize your trial wave function, by setting $dE(\beta)/d\beta = 0$. This tells you β as a function of γ (and hence—everything else being constant—of V_0), but let's use it instead to eliminate γ in favor of β:

$$E_{\min} = \frac{\hbar^2}{2\mu r_0^2}\frac{\beta^2(1-2\beta)}{(3+2\beta)}. \tag{8.87}$$

(d) Setting $\hbar^2/2\mu r_0^2 = 1$, plot $E_{\min}$ as a function of β, for $0 \le \beta \le 1$. What does this tell you about the binding of the deuteron? What is the minimum value of V_0 for which you can be confident there is a bound state (look up the necessary masses)? The experimental value is 52 MeV.

Problem 8.29 Existence of Bound States. A potential "well" (in one dimension) is a function $V(x)$ that is never positive ($V(x) \le 0$ for all x), and goes to zero at infinity ($V(x) \to 0$ as $x \to \pm\infty$).[24]

(a) Prove the following **Theorem:** If a potential well $V_1(x)$ supports at least one bound state, then any deeper/wider well ($V_2(x) \le V_1(x)$ for all x) will also support at least one bound state. *Hint:* Use the ground state of V_1, $\psi_1(x)$, as a variational test function.

(b) Prove the following **Corollary:** Every potential well in one dimension has a bound state.[25] *Hint:* Use a finite square well (Section 2.6) for V_1.

(c) Does the Theorem generalize to two and three dimensions? How about the Corollary? *Hint:* You might want to review Problems 4.11 and 4.51.

** **Problem 8.30** Performing a variational calculation requires finding the minimum of the energy, as a function of the variational parameters. This is, in general, a very hard problem. However, if we choose the form of our trial wave function judiciously, we can develop an efficient algorithm. In particular, suppose we use a *linear* combination of functions $\phi_n(x)$:

$$\psi(x) = \sum_{n=1}^{N} c_n\,\phi_n(x)\,, \tag{8.88}$$

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[24] To exclude trivial cases, we also assume it has nonzero area $\left(\int V(x)\,dx \ne 0\right)$. Notice that for the purposes of this problem neither the infinite square well nor the harmonic oscillator is a "potential well," though both of them, of course, have bound states.

[25] K. R. Brownstein, *Am. J. Phys.* **68**, 160 (2000) proves that any one-dimensional potential satisfying $\int_{-\infty}^{\infty} V(x)\,dx \le 0$ admits a bound state (as long as $V(x)$ is not identically zero)—even if it runs positive in some places.

where the c_n are the variational parameters. If the ϕ_n are an orthonormal set ($\langle \phi_m | \phi_n \rangle = \delta_{mn}$), but $\psi(x)$ is not necessarily normalized, then $\langle H \rangle$ is

$$\varepsilon = \frac{\langle \psi \mid H \mid \psi \rangle}{\langle \psi \mid \psi \rangle} = \frac{\sum_{mn} c_m^* \, H_{mn} \, c_n}{\sum_n |c_n|^2} \tag{8.89}$$

where $H_{mn} = \langle \phi_m | H | \phi_n \rangle$. Taking the derivative with respect to c_j^* (and setting the result equal to 0) gives[26]

$$\sum_n H_{jn} \, c_n = \varepsilon \, c_j \,, \tag{8.90}$$

recognizable as the jth row in an eigenvalue problem:

$$\begin{pmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ H_{21} & H_{22} & \cdots & H_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ H_{N1} & H_{N2} & \cdots & H_{NN} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} = \varepsilon \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}. \tag{8.91}$$

The smallest eigenvalue of this matrix H gives a bound on the ground state energy and the corresponding eigenvector determines the best variational wave function of the form 8.88.

(a) Verify Equation 8.90.

(b) Now take the derivative of Equation 8.89 with respect to c_j and show that you get a result redundant with Equation 8.90.

(c) Consider a particle in an infinite square well of width a, with a sloping floor:

$$V(x) = \begin{cases} \infty & x < 0, \\ V_0 \, x/a & 0 \le x \le a, \\ \infty & x > a \end{cases} .$$

Using a linear combination of the first ten stationary states of the infinite square well as the basis functions,

$$\phi_n = \sqrt{\frac{2}{a}} \, \sin\!\left(\frac{n \, \pi \, x}{a}\right) ,$$

determine a bound for the ground state energy in the case $V_0 = 100 \, \hbar^2/m \, a^2$. Make a plot of the optimized variational wave function. [*Note*: The exact result is $39.9819 \, \hbar^2/m \, a^2$.]

[26] Each c_j, being complex, stands for two independent parameters (its real and imaginary parts). One *could* take derivatives with respect to the real and imaginary parts,

$$\frac{\partial}{\partial \, \mathrm{Re}\left[c_j\right]} E = 0 \quad \text{and} \quad \frac{\partial}{\partial \, \mathrm{Im}\left[c_j\right]} E = 0,$$

but it is also legitimate (and simpler) to treat c_j and c_j^* as the independent parameters:

$$\frac{\partial}{\partial \, c_j} E = 0 \quad \text{and} \quad \frac{\partial}{\partial \, c_j^*} E = 0 \,.$$

You get the same result either way.